

Generative AI and its Interactions with the World of FOSS

Here's a primer on generative AI, thoughts on how you can empower yourself with this tech, and an insight into how Stable Diffusion has achieved success as an open source effort.



Figure 1: Three mysterious images

Take a look at Figure 1. Can you guess what these three images have in common? From a quick look, you may think that it's just the fantastical art style. What if I told you that none of them were drawn by a human artist but were generated on-the-fly within seconds by an AI model?

This is the power of generative AI — models capable of generating novel content based on a broad and abstract understanding of the world, driven by training over immense data sets. The 'artist' of the above images is a model named 'Stable Diffusion' released as open source software by the organisation Stability AI. All you have to give it is a text prompt describing the image you want in detail, and it will do the rest for you. If you're curious,

I've dropped example prompts that can generate the images in Figure 2.

Stable Diffusion is not the first of its kind. OpenAI's 'DALL-E' beat it to the chase as it was released a few months prior. However, looking at the state of the art, it is the first generative AI model capable of image synthesis and manipulation that was released to the public as FOSS software while also being practical to adopt.

These models are not limited to images. The first generation (and possibly the most popular use case) of these models targets text generation, and sometimes more specifically code generation. Recent advances even include speech and music creation. Zooming out into a timeline with some of the major generative AI milestones

(Figure 3), we see that this technology has been out for a while, but only gained monumental traction recently.

It all started with the paper titled 'Improving Language Understanding by Generative Pre-Training' released by OpenAI. The paper alluded to GPT (generative pretrained transformer), and laid the seed for 'large language models'. However, generative AI started coming into the limelight with the public release of GPT-3 and DALL-E. The former is a next-generation version of the already performant GPT-2 model and the latter is an image generative model like Stable Diffusion. Their public demos and API access set ablaze the imagination of technologists across the planet. However, these